Manish Jat

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SUMMARY

Backend Engineer with 1+ year of professional experience building Django and FastAPI applications. Experienced in designing production AI systems, distributed backend services, PostgreSQL databases, and Redis-based architecture. Built and deployed LLM-powered applications featuring vector search, streaming APIs, distributed rate limiting, and optimized ML pipelines.

EXPERIENCE

Django Developer Aug 2025 – Feb 2026

JJM Software *Bhilwara*

- Maintained and enhanced backend systems across 5 production applications using Django and Python.
- Designed and deployed 8+ REST APIs serving client-facing applications.
- Optimized database queries and indexing strategies, reducing API latency by 30%.
- Resolved 70+ critical bugs and feature requests across multiple projects.
- Contributed to 700+ development tickets including production support, bug fixes, and enhancements.

PROJECTS

DocMind — Production RAG Platform | FastAPI • Gemini • Pinecone • PostgreSQL • Redis • Cloud Run

- Built a multi-tenant document Q&A platform supporting PDF and CSV ingestion with citation-based responses.
- Implemented streaming AI responses using Server-Sent Events.
- Designed an 18-key Gemini API pool with automatic rate-limit handling and failover.
- Integrated vector search using Pinecone namespaces for tenant isolation.
- Deployed production infrastructure using Cloud Run, PostgreSQL, and Redis.

AI Coding Agent | Gemini API • FastAPI • WebSockets • VS Code Extension

- Built a Claude-Code style AI coding assistant with planner, critic, and reviewer agent architecture.
- Implemented 48+ tools including file operations, shell execution, Git integration, browser automation, and AST analysis.
- Added snapshot and rollback systems for safe code modification workflows.
- Developed WebSocket-based real-time communication and multi-platform support (Web, CLI, VS Code).

Distributed API Rate Limiter | FastAPI • Redis Lua • PostgreSQL

- Implemented sliding-window, token-bucket, fixed-window, and leaky-bucket rate-limiting algorithms.
- Used atomic Redis Lua scripts to eliminate race conditions in distributed environments.
- Built anomaly detection using Isolation Forest to identify suspicious traffic patterns.
- Designed Redis hot-cache and PostgreSQL persistence architecture for scalability.

SentinelLM | DistilBERT • ONNX • FastAPI • Redis

- Fine-tuned a transformer model on 200k+ records for toxicity classification.
- Reduced CPU inference latency from 24.8ms to 9.7ms through ONNX optimization.
- Added Redis caching and rule-based shortcuts for low-latency predictions.
- Built production APIs with testing, logging, and model versioning workflows.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Bash, Lua

Backend: FastAPI, Django, Flask, REST APIs, Async Programming, WebSockets, JWT Authentication

Databases: PostgreSQL, Redis, TimescaleDB, Pinecone (Vector Database)

AI/LLM: RAG, MCP, Agentic Workflows, LangChain, DistilBERT, ONNX, MLflow, Sentence Transformers

Distributed Systems: Redis Lua Scripts, Atomic Operations, Caching, Rate Limiting, Anomaly Detection

Cloud & DevOps: Docker, Google Cloud Run, Git, GitHub Actions, Cloud Scheduler, Vercel, Linux

Vector Search: Vector Search, Embeddings, Database Indexing, Query Optimization

EDUCATION

Indian Institute of Technology Madras

B.S. in Data Science and Applications, CGPA: 7.8 *2023 – 2026*

SVGMS, Bhilwara

Senior Secondary (PCM), 79.8% *2020 – 2022*